MSG No. 219.88 $F_S\bar{4}3c$ [Type IV, cubic]

Table 1: Wyckoff site: `8a`, site symmetry: `-4'3m'`

No.	position	mapping
1	$[0, 0, 0]$	[1,2,3,4,5,6,7,8,9,10,11,12,181,182,183,184,185,186,187,188,189,190,191,192]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[13,14,15,16,17,18,19,20,21,22,23,24,169,170,171,172,173,174,175,176,177,178,179,180]
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[25,26,27,28,29,30,31,32,33,34,35,36,157,158,159,160,161,162,163,164,165,166,167,168]
4	$\left[\frac{1}{2}, 0, 0\right]$	[37,38,39,40,41,42,43,44,45,46,47,48,145,146,147,148,149,150,151,152,153,154,155,156]
5	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[49,50,51,52,53,54,55,56,57,58,59,60,133,134,135,136,137,138,139,140,141,142,143,144]
6	$\left[0, \frac{1}{2}, 0\right]$	[61,62,63,64,65,66,67,68,69,70,71,72,121,122,123,124,125,126,127,128,129,130,131,132]
7	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[73,74,75,76,77,78,79,80,81,82,83,84,109,110,111,112,113,114,115,116,117,118,119,120]
8	$\left[0, 0, \frac{1}{2}\right]$	[85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100,101,102,103,104,105,106,107,108]

Table 2: Wyckoff site: `8b`, site symmetry: `-4'3m'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,5,6,26,34,35,51,55,60,76,80,81,110,111,115,133,138,143,160,161,165,188,190,192]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[2,10,11,25,29,30,52,56,57,75,79,84,109,114,119,134,135,139,164,166,168,184,185,189]
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[3,7,12,28,32,33,49,53,54,74,82,83,112,113,117,140,142,144,158,159,163,181,186,191]
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[4,8,9,27,31,36,50,58,59,73,77,78,116,118,120,136,137,141,157,162,167,182,183,187]
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[13,18,23,38,39,43,68,70,72,88,89,93,98,106,107,121,125,126,148,152,153,171,175,180]
6	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[14,15,19,37,42,47,64,65,69,92,94,96,97,101,102,122,130,131,147,151,156,172,176,177]
7	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[16,17,21,44,46,48,62,63,67,85,90,95,99,103,108,124,128,129,145,149,150,170,178,179]
8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[20,22,24,40,41,45,61,66,71,86,87,91,100,104,105,123,127,132,146,154,155,169,173,174]

Table 3: Wyckoff site: `24c`, site symmetry: `-42'.m'`

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	[1,26,62,85,99,124,165,190]
2	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	[2,25,61,86,100,123,166,189]
3	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[3,28,69,94,97,122,158,181]
4	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[4,27,70,93,98,121,157,182]
5	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	[5,43,60,89,106,138,152,188]
6	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	[6,47,64,81,111,131,151,192]
7	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[7,48,63,83,112,129,150,191]
8	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	[8,44,58,90,108,137,149,187]
9	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	[9,40,71,78,120,127,155,183]
10	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	[10,42,56,92,101,139,156,185]
11	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[11,39,72,79,119,126,153,184]
12	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	[12,41,53,91,104,140,154,186]
13	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[13,38,50,73,118,141,148,171]
14	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[14,37,49,74,117,142,147,172]
15	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[15,35,55,96,102,143,160,177]
16	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[16,33,54,95,103,144,159,179]
17	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[17,36,67,77,116,128,162,178]
18	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[18,34,68,80,115,125,161,180]

continued ...

Table 3

No.	position	mapping
19	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[19,29,65,84,114,130,164,176]`
20	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[20,32,66,82,113,132,163,173]`
21	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	`[21,46,51,76,110,133,145,170]`
22	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[22,45,52,75,109,134,146,169]`
23	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[23,30,57,88,107,135,168,175]`
24	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[24,31,59,87,105,136,167,174]`

Table 4: Wyckoff site: `24d`, site symmetry: `-42'.m'`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, 0\right]$	`[1,2,37,38,147,148,189,190]`
2	$\left[\frac{3}{4}, 0, 0\right]$	`[3,4,45,46,145,146,181,182]`
3	$\left[0, \frac{1}{4}, 0\right]$	`[5,12,65,67,128,130,186,188]`
4	$\left[0, 0, \frac{1}{4}\right]$	`[6,9,88,95,103,107,183,192]`
5	$\left[0, 0, \frac{3}{4}\right]$	`[7,11,87,96,102,105,184,191]`
6	$\left[0, \frac{3}{4}, 0\right]$	`[8,10,66,68,125,132,185,187]`
7	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[13,14,25,26,165,166,171,172]`
8	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[15,24,79,83,112,119,174,177]`
9	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[16,23,78,81,111,120,175,179]`
10	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}\right]$	`[17,19,53,60,138,140,176,178]`
11	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{2}\right]$	`[18,20,56,58,137,139,173,180]`
12	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[21,22,27,28,157,158,169,170]`
13	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[29,36,89,91,104,106,162,164]`
14	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[30,33,64,71,127,131,159,168]`
15	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[31,35,63,72,126,129,160,167]`
16	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[32,34,90,92,101,108,161,163]`
17	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[39,48,55,59,136,143,150,153]`
18	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[40,47,54,57,135,144,151,155]`
19	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[41,43,77,84,114,116,152,154]`
20	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[42,44,80,82,113,115,149,156]`
21	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[49,50,85,86,99,100,141,142]`
22	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[51,52,93,94,97,98,133,134]`
23	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[61,62,73,74,117,118,123,124]`
24	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[69,70,75,76,109,110,121,122]`

Table 5: Wyckoff site: `32e`, site symmetry: `.3m'`

No.	position	mapping
1	$[x, x, x]$	`[1,5,6,188,190,192]`
2	$[x, -x, -x]$	`[2,10,11,184,185,189]`
3	$[-x, x, -x]$	`[3,7,12,181,186,191]`
4	$[-x, -x, x]$	`[4,8,9,182,183,187]`
5	$\left[\frac{1}{2} - x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	`[13,18,23,171,175,180]`
6	$\left[\frac{1}{2} - x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	`[14,15,19,172,176,177]`

continued ...

Table 5

No.	position	mapping
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	[16,17,21,170,178,179]
8	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	[20,22,24,169,173,174]
9	$\left[x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	[25,29,30,164,166,168]
10	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	[26,34,35,160,161,165]
11	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	[27,31,36,157,162,167]
12	$\left[-x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	[28,32,33,158,159,163]
13	$\left[\frac{1}{2}-x,\ x,\ -x\right]$	[37,42,47,147,151,156]
14	$\left[\frac{1}{2}-x,\ -x,\ x\right]$	[38,39,43,148,152,153]
15	$\left[x+\frac{1}{2},\ -x,\ -x\right]$	[40,41,45,146,154,155]
16	$\left[x+\frac{1}{2},\ x,\ x\right]$	[44,46,48,145,149,150]
17	$\left[x+\frac{1}{2},\ x,\ x+\frac{1}{2}\right]$	[49,53,54,140,142,144]
18	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{2}-x\right]$	[50,58,59,136,137,141]
19	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}-x\right]$	[51,55,60,133,138,143]
20	$\left[\frac{1}{2}-x,\ -x,\ x+\frac{1}{2}\right]$	[52,56,57,134,135,139]
21	$\left[-x,\ x+\frac{1}{2},\ -x\right]$	[61,66,71,123,127,132]
22	$\left[-x,\ \frac{1}{2}-x,\ x\right]$	[62,63,67,124,128,129]
23	$\left[x,\ \frac{1}{2}-x,\ -x\right]$	[64,65,69,122,130,131]
24	$\left[x,\ x+\frac{1}{2},\ x\right]$	[68,70,72,121,125,126]
25	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x\right]$	[73,77,78,116,118,120]
26	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -x\right]$	[74,82,83,112,113,117]
27	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ -x\right]$	[75,79,84,109,114,119]
28	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x\right]$	[76,80,81,110,111,115]
29	$\left[-x,\ x,\ \frac{1}{2}-x\right]$	[85,90,95,99,103,108]
30	$\left[-x,\ -x,\ x+\frac{1}{2}\right]$	[86,87,91,100,104,105]
31	$\left[x,\ -x,\ \frac{1}{2}-x\right]$	[88,89,93,98,106,107]
32	$\left[x,\ x,\ x+\frac{1}{2}\right]$	[92,94,96,97,101,102]

Table 6: Wyckoff site: 48f, site symmetry: 2.m'm'

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,2,189,190]
2	$[-x,\ 0,\ 0]$	[3,4,181,182]
3	$[0,\ x,\ 0]$	[5,12,186,188]
4	$[0,\ 0,\ x]$	[6,9,183,192]
5	$[0,\ 0,\ -x]$	[7,11,184,191]
6	$[0,\ -x,\ 0]$	[8,10,185,187]
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[13,14,171,172]
8	$\left[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	[15,24,174,177]
9	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	[16,23,175,179]
10	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[17,19,176,178]
11	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[18,20,173,180]
12	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[21,22,169,170]
13	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[25,26,165,166]
14	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[27,28,157,158]
15	$\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[29,36,162,164]

continued ...

Table 6

No.	position	mapping
16	$\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$	[30,33,159,168]
17	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$	[31,35,160,167]
18	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[32,34,161,163]
19	$\left[\frac{1}{2}-x,\ 0,\ 0\right]$	[37,38,147,148]
20	$\left[\frac{1}{2},\ 0,\ x\right]$	[39,48,150,153]
21	$\left[\frac{1}{2},\ 0,\ -x\right]$	[40,47,151,155]
22	$\left[\frac{1}{2},\ -x,\ 0\right]$	[41,43,152,154]
23	$\left[\frac{1}{2},\ x,\ 0\right]$	[42,44,149,156]
24	$\left[x+\frac{1}{2},\ 0,\ 0\right]$	[45,46,145,146]
25	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[49,50,141,142]
26	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[51,52,133,134]
27	$\left[\frac{1}{2},\ x,\ \frac{1}{2}\right]$	[53,60,138,140]
28	$\left[\frac{1}{2},\ 0,\ x+\frac{1}{2}\right]$	[54,57,135,144]
29	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-x\right]$	[55,59,136,143]
30	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	[56,58,137,139]
31	$\left[-x,\ \frac{1}{2},\ 0\right]$	[61,62,123,124]
32	$\left[0,\ \frac{1}{2},\ x\right]$	[63,72,126,129]
33	$\left[0,\ \frac{1}{2},\ -x\right]$	[64,71,127,131]
34	$\left[0,\ \frac{1}{2}-x,\ 0\right]$	[65,67,128,130]
35	$\left[0,\ x+\frac{1}{2},\ 0\right]$	[66,68,125,132]
36	$\left[x,\ \frac{1}{2},\ 0\right]$	[69,70,121,122]
37	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[73,74,117,118]
38	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	[75,76,109,110]
39	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[77,84,114,116]
40	$\left[\frac{1}{2},\ \frac{1}{2},\ x\right]$	[78,81,111,120]
41	$\left[\frac{1}{2},\ \frac{1}{2},\ -x\right]$	[79,83,112,119]
42	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	[80,82,113,115]
43	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[85,86,99,100]
44	$\left[0,\ 0,\ x+\frac{1}{2}\right]$	[87,96,102,105]
45	$\left[0,\ 0,\ \frac{1}{2}-x\right]$	[88,95,103,107]
46	$\left[0,\ -x,\ \frac{1}{2}\right]$	[89,91,104,106]
47	$\left[0,\ x,\ \frac{1}{2}\right]$	[90,92,101,108]
48	$\left[x,\ 0,\ \frac{1}{2}\right]$	[93,94,97,98]

Table 7: Wyckoff site: 48g, site symmetry: 2.m'm'

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	[1,26,165,190]
2	$\left[x,\ \frac{3}{4},\ \frac{3}{4}\right]$	[2,25,166,189]
3	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[3,28,158,181]
4	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[4,27,157,182]
5	$\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$	[5,60,138,188]
6	$\left[\frac{1}{4},\ \frac{1}{4},\ x\right]$	[6,81,111,192]
7	$\left[\frac{3}{4},\ \frac{1}{4},\ -x\right]$	[7,83,112,191]
8	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	[8,58,137,187]

continued ...

Table 7

No.	position	mapping
9	$\left[\frac{3}{4}, \frac{3}{4}, x\right]$	[9,78,120,183]
10	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[10,56,139,185]
11	$\left[\frac{1}{4}, \frac{3}{4}, -x\right]$	[11,79,119,184]
12	$\left[\frac{3}{4}, x, \frac{3}{4}\right]$	[12,53,140,186]
13	$\left[\frac{1}{2} - x, \frac{3}{4}, \frac{1}{4}\right]$	[13,38,148,171]
14	$\left[\frac{1}{2} - x, \frac{1}{4}, \frac{3}{4}\right]$	[14,37,147,172]
15	$\left[\frac{1}{4}, \frac{1}{4}, x + \frac{1}{2}\right]$	[15,96,102,177]
16	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2} - x\right]$	[16,95,103,179]
17	$\left[\frac{3}{4}, \frac{1}{2} - x, \frac{1}{4}\right]$	[17,67,128,178]
18	$\left[\frac{1}{4}, x + \frac{1}{2}, \frac{1}{4}\right]$	[18,68,125,180]
19	$\left[\frac{1}{4}, \frac{1}{2} - x, \frac{3}{4}\right]$	[19,65,130,176]
20	$\left[\frac{3}{4}, x + \frac{1}{2}, \frac{3}{4}\right]$	[20,66,132,173]
21	$\left[x + \frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	[21,46,145,170]
22	$\left[x + \frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[22,45,146,169]
23	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2} - x\right]$	[23,88,107,175]
24	$\left[\frac{3}{4}, \frac{3}{4}, x + \frac{1}{2}\right]$	[24,87,105,174]
25	$\left[\frac{1}{4}, x + \frac{1}{2}, \frac{3}{4}\right]$	[29,84,114,164]
26	$\left[\frac{1}{4}, \frac{3}{4}, x + \frac{1}{2}\right]$	[30,57,135,168]
27	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2} - x\right]$	[31,59,136,167]
28	$\left[\frac{3}{4}, \frac{1}{2} - x, \frac{3}{4}\right]$	[32,82,113,163]
29	$\left[\frac{3}{4}, \frac{1}{4}, x + \frac{1}{2}\right]$	[33,54,144,159]
30	$\left[\frac{1}{4}, \frac{1}{2} - x, \frac{1}{4}\right]$	[34,80,115,161]
31	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - x\right]$	[35,55,143,160]
32	$\left[\frac{3}{4}, x + \frac{1}{2}, \frac{1}{4}\right]$	[36,77,116,162]
33	$\left[\frac{1}{4}, \frac{3}{4}, x\right]$	[39,72,126,153]
34	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[40,71,127,155]
35	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[41,91,104,154]
36	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	[42,92,101,156]
37	$\left[\frac{1}{4}, -x, \frac{1}{4}\right]$	[43,89,106,152]
38	$\left[\frac{3}{4}, x, \frac{1}{4}\right]$	[44,90,108,149]
39	$\left[\frac{1}{4}, \frac{1}{4}, -x\right]$	[47,64,131,151]
40	$\left[\frac{3}{4}, \frac{1}{4}, x\right]$	[48,63,129,150]
41	$\left[x + \frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[49,74,117,142]
42	$\left[x + \frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[50,73,118,141]
43	$\left[\frac{1}{2} - x, \frac{1}{4}, \frac{1}{4}\right]$	[51,76,110,133]
44	$\left[\frac{1}{2} - x, \frac{3}{4}, \frac{3}{4}\right]$	[52,75,109,134]
45	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[61,86,100,123]
46	$\left[-x, \frac{1}{4}, \frac{1}{4}\right]$	[62,85,99,124]
47	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	[69,94,97,122]
48	$\left[x, \frac{3}{4}, \frac{1}{4}\right]$	[70,93,98,121]

Table 8: Wyckoff site: 96h, site symmetry: 2'..

No.	position	mapping
1	$\left[x, \frac{1}{4}, 0\right]$	[1,122]

continued ...

Table 8

No.	position	mapping
2	$\left[x,\ \frac{3}{4},\ 0\right]$	[2,121]
3	$\left[-x,\ \frac{1}{4},\ 0\right]$	[3,124]
4	$\left[-x,\ \frac{3}{4},\ 0\right]$	[4,123]
5	$\left[0,\ x,\ \frac{1}{4}\right]$	[5,108]
6	$\left[\frac{1}{4},\ 0,\ x\right]$	[6,153]
7	$\left[\frac{3}{4},\ 0,\ -x\right]$	[7,155]
8	$\left[0,\ -x,\ \frac{1}{4}\right]$	[8,106]
9	$\left[\frac{3}{4},\ 0,\ x\right]$	[9,150]
10	$\left[0,\ -x,\ \frac{3}{4}\right]$	[10,104]
11	$\left[\frac{1}{4},\ 0,\ -x\right]$	[11,151]
12	$\left[0,\ x,\ \frac{3}{4}\right]$	[12,101]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[13,110]
14	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[14,109]
15	$\left[\frac{1}{2},\ \frac{1}{4},\ x+\frac{1}{2}\right]$	[15,144]
16	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	[16,143]
17	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[17,163]
18	$\left[\frac{1}{4},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[18,164]
19	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[19,161]
20	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[20,162]
21	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[21,118]
22	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[22,117]
23	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}-x\right]$	[23,136]
24	$\left[\frac{1}{2},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	[24,135]
25	$\left[x,\ \frac{3}{4},\ \frac{1}{2}\right]$	[25,98]
26	$\left[x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[26,97]
27	$\left[-x,\ \frac{3}{4},\ \frac{1}{2}\right]$	[27,100]
28	$\left[-x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[28,99]
29	$\left[0,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[29,132]
30	$\left[\frac{1}{4},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	[30,177]
31	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	[31,179]
32	$\left[0,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[32,130]
33	$\left[\frac{3}{4},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	[33,174]
34	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[34,128]
35	$\left[\frac{1}{4},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	[35,175]
36	$\left[0,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[36,125]
37	$\left[\frac{1}{2}-x,\ 0,\ \frac{3}{4}\right]$	[37,134]
38	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{4}\right]$	[38,133]
39	$\left[\frac{1}{2},\ \frac{3}{4},\ x\right]$	[39,120]
40	$\left[\frac{1}{2},\ \frac{3}{4},\ -x\right]$	[40,119]
41	$\left[\frac{3}{4},\ -x,\ 0\right]$	[41,187]
42	$\left[\frac{1}{4},\ x,\ 0\right]$	[42,188]
43	$\left[\frac{1}{4},\ -x,\ 0\right]$	[43,185]
44	$\left[\frac{3}{4},\ x,\ 0\right]$	[44,186]
45	$\left[x+\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	[45,142]
46	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	[46,141]
47	$\left[\frac{1}{2},\ \frac{1}{4},\ -x\right]$	[47,112]
48	$\left[\frac{1}{2},\ \frac{1}{4},\ x\right]$	[48,111]

continued ...

Table 8

No.	position	mapping
49	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{1}{2}\right]$	[49,170]
50	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{2}\right]$	[50,169]
51	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{2}\right]$	[51,172]
52	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{2}\right]$	[52,171]
53	$\left[\frac{1}{2}, x, \frac{3}{4}\right]$	[53,156]
54	$\left[\frac{3}{4}, 0, x+\frac{1}{2}\right]$	[54,105]
55	$\left[\frac{1}{4}, 0, \frac{1}{2}-x\right]$	[55,107]
56	$\left[\frac{1}{2}, -x, \frac{3}{4}\right]$	[56,154]
57	$\left[\frac{1}{4}, 0, x+\frac{1}{2}\right]$	[57,102]
58	$\left[\frac{1}{2}, -x, \frac{1}{4}\right]$	[58,152]
59	$\left[\frac{3}{4}, 0, \frac{1}{2}-x\right]$	[59,103]
60	$\left[\frac{1}{2}, x, \frac{1}{4}\right]$	[60,149]
61	$\left[-x, \frac{1}{2}, \frac{3}{4}\right]$	[61,158]
62	$\left[-x, \frac{1}{2}, \frac{1}{4}\right]$	[62,157]
63	$\left[0, \frac{1}{4}, x\right]$	[63,192]
64	$\left[0, \frac{1}{4}, -x\right]$	[64,191]
65	$\left[\frac{1}{4}, \frac{1}{2}-x, 0\right]$	[65,115]
66	$\left[\frac{3}{4}, x+\frac{1}{2}, 0\right]$	[66,116]
67	$\left[\frac{3}{4}, \frac{1}{2}-x, 0\right]$	[67,113]
68	$\left[\frac{1}{4}, x+\frac{1}{2}, 0\right]$	[68,114]
69	$\left[x, \frac{1}{2}, \frac{3}{4}\right]$	[69,166]
70	$\left[x, \frac{1}{2}, \frac{1}{4}\right]$	[70,165]
71	$\left[0, \frac{3}{4}, -x\right]$	[71,184]
72	$\left[0, \frac{3}{4}, x\right]$	[72,183]
73	$\left[x+\frac{1}{2}, \frac{3}{4}, 0\right]$	[73,146]
74	$\left[x+\frac{1}{2}, \frac{1}{4}, 0\right]$	[74,145]
75	$\left[\frac{1}{2}-x, \frac{3}{4}, 0\right]$	[75,148]
76	$\left[\frac{1}{2}-x, \frac{1}{4}, 0\right]$	[76,147]
77	$\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{4}\right]$	[77,180]
78	$\left[\frac{3}{4}, \frac{1}{2}, x\right]$	[78,129]
79	$\left[\frac{1}{4}, \frac{1}{2}, -x\right]$	[79,131]
80	$\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{4}\right]$	[80,178]
81	$\left[\frac{1}{4}, \frac{1}{2}, x\right]$	[81,126]
82	$\left[\frac{1}{2}, \frac{1}{2}-x, \frac{3}{4}\right]$	[82,176]
83	$\left[\frac{3}{4}, \frac{1}{2}, -x\right]$	[83,127]
84	$\left[\frac{1}{2}, x+\frac{1}{2}, \frac{3}{4}\right]$	[84,173]
85	$\left[-x, 0, \frac{1}{4}\right]$	[85,182]
86	$\left[-x, 0, \frac{3}{4}\right]$	[86,181]
87	$\left[0, \frac{3}{4}, x+\frac{1}{2}\right]$	[87,168]
88	$\left[0, \frac{3}{4}, \frac{1}{2}-x\right]$	[88,167]
89	$\left[\frac{1}{4}, -x, \frac{1}{2}\right]$	[89,139]
90	$\left[\frac{3}{4}, x, \frac{1}{2}\right]$	[90,140]
91	$\left[\frac{3}{4}, -x, \frac{1}{2}\right]$	[91,137]
92	$\left[\frac{1}{4}, x, \frac{1}{2}\right]$	[92,138]
93	$\left[x, 0, \frac{1}{4}\right]$	[93,190]
94	$\left[x, 0, \frac{3}{4}\right]$	[94,189]
95	$\left[0, \frac{1}{4}, \frac{1}{2}-x\right]$	[95,160]

continued ...

Table 8

No.	position	mapping
96	$\left[0,\, \frac{1}{4},\, x+\frac{1}{2}\right]$	[96,159]

Table 9: Wyckoff site: 96i, site symmetry: ..m'

No.	position	mapping
1	$[x,\, x,\, z]$	[1,188]
2	$[x,\, -x,\, -z]$	[2,185]
3	$[-x,\, x,\, -z]$	[3,186]
4	$[-x,\, -x,\, z]$	[4,187]
5	$[z,\, x,\, x]$	[5,192]
6	$[x,\, z,\, x]$	[6,190]
7	$[-x,\, z,\, -x]$	[7,181]
8	$[-z,\, -x,\, x]$	[8,183]
9	$[-x,\, -z,\, x]$	[9,182]
10	$[z,\, -x,\, -x]$	[10,184]
11	$[x,\, -z,\, -x]$	[11,189]
12	$[-z,\, x,\, -x]$	[12,191]
13	$\left[\frac{1}{2}-x,\, z+\frac{1}{2},\, \frac{1}{2}-x\right]$	[13,175]
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-z,\, x+\frac{1}{2}\right]$	[14,177]
15	$\left[\frac{1}{2}-z,\, \frac{1}{2}-x,\, x+\frac{1}{2}\right]$	[15,176]
16	$\left[z+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-x\right]$	[16,178]
17	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[17,170]
18	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[18,171]
19	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[19,172]
20	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[20,169]
21	$\left[x+\frac{1}{2},\, \frac{1}{2}-z,\, \frac{1}{2}-x\right]$	[21,179]
22	$\left[x+\frac{1}{2},\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$	[22,174]
23	$\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, \frac{1}{2}-x\right]$	[23,180]
24	$\left[z+\frac{1}{2},\, x+\frac{1}{2},\, x+\frac{1}{2}\right]$	[24,173]
25	$\left[x,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[25,164]
26	$\left[x,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[26,161]
27	$\left[-x,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[27,162]
28	$\left[-x,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[28,163]
29	$\left[z,\, x+\frac{1}{2},\, x+\frac{1}{2}\right]$	[29,168]
30	$\left[x,\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$	[30,166]
31	$\left[-x,\, z+\frac{1}{2},\, \frac{1}{2}-x\right]$	[31,157]
32	$\left[-z,\, \frac{1}{2}-x,\, x+\frac{1}{2}\right]$	[32,159]
33	$\left[-x,\, \frac{1}{2}-z,\, x+\frac{1}{2}\right]$	[33,158]
34	$\left[z,\, \frac{1}{2}-x,\, \frac{1}{2}-x\right]$	[34,160]
35	$\left[x,\, \frac{1}{2}-z,\, \frac{1}{2}-x\right]$	[35,165]
36	$\left[-z,\, x+\frac{1}{2},\, \frac{1}{2}-x\right]$	[36,167]
37	$\left[\frac{1}{2}-x,\, z,\, -x\right]$	[37,151]
38	$\left[\frac{1}{2}-x,\, -z,\, x\right]$	[38,153]
39	$\left[\frac{1}{2}-z,\, -x,\, x\right]$	[39,152]
40	$\left[z+\frac{1}{2},\, -x,\, -x\right]$	[40,154]

continued ...

Table 9

No.	position	mapping
41	$\left[x+\frac{1}{2},\,-x,\,-z\right]$	[41,146]
42	$\left[\frac{1}{2}-x,\,x,\,-z\right]$	[42,147]
43	$\left[\frac{1}{2}-x,\,-x,\,z\right]$	[43,148]
44	$\left[x+\frac{1}{2},\,x,\,z\right]$	[44,145]
45	$\left[x+\frac{1}{2},\,-z,\,-x\right]$	[45,155]
46	$\left[x+\frac{1}{2},\,z,\,x\right]$	[46,150]
47	$\left[\frac{1}{2}-z,\,x,\,-x\right]$	[47,156]
48	$\left[z+\frac{1}{2},\,x,\,x\right]$	[48,149]
49	$\left[x+\frac{1}{2},\,x,\,z+\frac{1}{2}\right]$	[49,140]
50	$\left[x+\frac{1}{2},\,-x,\,\frac{1}{2}-z\right]$	[50,137]
51	$\left[\frac{1}{2}-x,\,x,\,\frac{1}{2}-z\right]$	[51,138]
52	$\left[\frac{1}{2}-x,\,-x,\,z+\frac{1}{2}\right]$	[52,139]
53	$\left[z+\frac{1}{2},\,x,\,x+\frac{1}{2}\right]$	[53,144]
54	$\left[x+\frac{1}{2},\,z,\,x+\frac{1}{2}\right]$	[54,142]
55	$\left[\frac{1}{2}-x,\,z,\,\frac{1}{2}-x\right]$	[55,133]
56	$\left[\frac{1}{2}-z,\,-x,\,x+\frac{1}{2}\right]$	[56,135]
57	$\left[\frac{1}{2}-x,\,-z,\,x+\frac{1}{2}\right]$	[57,134]
58	$\left[z+\frac{1}{2},\,-x,\,\frac{1}{2}-x\right]$	[58,136]
59	$\left[x+\frac{1}{2},\,-z,\,\frac{1}{2}-x\right]$	[59,141]
60	$\left[\frac{1}{2}-z,\,x,\,\frac{1}{2}-x\right]$	[60,143]
61	$\left[-x,\,z+\frac{1}{2},\,-x\right]$	[61,127]
62	$\left[-x,\,\frac{1}{2}-z,\,x\right]$	[62,129]
63	$\left[-z,\,\frac{1}{2}-x,\,x\right]$	[63,128]
64	$\left[z,\,\frac{1}{2}-x,\,-x\right]$	[64,130]
65	$\left[x,\,\frac{1}{2}-x,\,-z\right]$	[65,122]
66	$\left[-x,\,x+\frac{1}{2},\,-z\right]$	[66,123]
67	$\left[-x,\,\frac{1}{2}-x,\,z\right]$	[67,124]
68	$\left[x,\,x+\frac{1}{2},\,z\right]$	[68,121]
69	$\left[x,\,\frac{1}{2}-z,\,-x\right]$	[69,131]
70	$\left[x,\,z+\frac{1}{2},\,x\right]$	[70,126]
71	$\left[-z,\,x+\frac{1}{2},\,-x\right]$	[71,132]
72	$\left[z,\,x+\frac{1}{2},\,x\right]$	[72,125]
73	$\left[x+\frac{1}{2},\,x+\frac{1}{2},\,z\right]$	[73,116]
74	$\left[x+\frac{1}{2},\,\frac{1}{2}-x,\,-z\right]$	[74,113]
75	$\left[\frac{1}{2}-x,\,x+\frac{1}{2},\,-z\right]$	[75,114]
76	$\left[\frac{1}{2}-x,\,\frac{1}{2}-x,\,z\right]$	[76,115]
77	$\left[z+\frac{1}{2},\,x+\frac{1}{2},\,x\right]$	[77,120]
78	$\left[x+\frac{1}{2},\,z+\frac{1}{2},\,x\right]$	[78,118]
79	$\left[\frac{1}{2}-x,\,z+\frac{1}{2},\,-x\right]$	[79,109]
80	$\left[\frac{1}{2}-z,\,\frac{1}{2}-x,\,x\right]$	[80,111]
81	$\left[\frac{1}{2}-x,\,\frac{1}{2}-z,\,x\right]$	[81,110]
82	$\left[z+\frac{1}{2},\,\frac{1}{2}-x,\,-x\right]$	[82,112]
83	$\left[x+\frac{1}{2},\,\frac{1}{2}-z,\,-x\right]$	[83,117]
84	$\left[\frac{1}{2}-z,\,x+\frac{1}{2},\,-x\right]$	[84,119]
85	$\left[-x,\,z,\,\frac{1}{2}-x\right]$	[85,103]
86	$\left[-x,\,-z,\,x+\frac{1}{2}\right]$	[86,105]
87	$\left[-z,\,-x,\,x+\frac{1}{2}\right]$	[87,104]

continued ...

Table 9

No.	position	mapping
88	$\left[z,\, -x,\, \frac{1}{2}-x\right]$	[88,106]
89	$\left[x,\, -x,\, \frac{1}{2}-z\right]$	[89,98]
90	$\left[-x,\, x,\, \frac{1}{2}-z\right]$	[90,99]
91	$\left[-x,\, -x,\, z+\frac{1}{2}\right]$	[91,100]
92	$\left[x,\, x,\, z+\frac{1}{2}\right]$	[92,97]
93	$\left[x,\, -z,\, \frac{1}{2}-x\right]$	[93,107]
94	$\left[x,\, z,\, x+\frac{1}{2}\right]$	[94,102]
95	$\left[-z,\, x,\, \frac{1}{2}-x\right]$	[95,108]
96	$\left[z,\, x,\, x+\frac{1}{2}\right]$	[96,101]

Table 10: Wyckoff site: 192j, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[x,\, -y,\, -z]$	[2]
3	$[-x,\, y,\, -z]$	[3]
4	$[-x,\, -y,\, z]$	[4]
5	$[z,\, x,\, y]$	[5]
6	$[y,\, z,\, x]$	[6]
7	$[-y,\, z,\, -x]$	[7]
8	$[-z,\, -x,\, y]$	[8]
9	$[-y,\, -z,\, x]$	[9]
10	$[z,\, -x,\, -y]$	[10]
11	$[y,\, -z,\, -x]$	[11]
12	$[-z,\, x,\, -y]$	[12]
13	$\left[\frac{1}{2}-x,\, z+\frac{1}{2},\, \frac{1}{2}-y\right]$	[13]
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-z,\, y+\frac{1}{2}\right]$	[14]
15	$\left[\frac{1}{2}-z,\, \frac{1}{2}-y,\, x+\frac{1}{2}\right]$	[15]
16	$\left[z+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-x\right]$	[16]
17	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[17]
18	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[18]
19	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[19]
20	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[20]
21	$\left[x+\frac{1}{2},\, \frac{1}{2}-z,\, \frac{1}{2}-y\right]$	[21]
22	$\left[x+\frac{1}{2},\, z+\frac{1}{2},\, y+\frac{1}{2}\right]$	[22]
23	$\left[\frac{1}{2}-z,\, y+\frac{1}{2},\, \frac{1}{2}-x\right]$	[23]
24	$\left[z+\frac{1}{2},\, y+\frac{1}{2},\, x+\frac{1}{2}\right]$	[24]
25	$\left[x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[25]
26	$\left[x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[26]
27	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[27]
28	$\left[-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[28]
29	$\left[z,\, x+\frac{1}{2},\, y+\frac{1}{2}\right]$	[29]
30	$\left[y,\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$	[30]
31	$\left[-y,\, z+\frac{1}{2},\, \frac{1}{2}-x\right]$	[31]
32	$\left[-z,\, \frac{1}{2}-x,\, y+\frac{1}{2}\right]$	[32]

continued ...

Table 10

No.	position	mapping
33	$\left[-y,\ \frac{1}{2}-z,\ x+\frac{1}{2}\right]$	[33]
34	$\left[z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$	[34]
35	$\left[y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[35]
36	$\left[-z,\ x+\frac{1}{2},\ \frac{1}{2}-y\right]$	[36]
37	$\left[\frac{1}{2}-x,\ z,\ -y\right]$	[37]
38	$\left[\frac{1}{2}-x,\ -z,\ y\right]$	[38]
39	$\left[\frac{1}{2}-z,\ -y,\ x\right]$	[39]
40	$\left[z+\frac{1}{2},\ -y,\ -x\right]$	[40]
41	$\left[y+\frac{1}{2},\ -x,\ -z\right]$	[41]
42	$\left[\frac{1}{2}-y,\ x,\ -z\right]$	[42]
43	$\left[\frac{1}{2}-y,\ -x,\ z\right]$	[43]
44	$\left[y+\frac{1}{2},\ x,\ z\right]$	[44]
45	$\left[x+\frac{1}{2},\ -z,\ -y\right]$	[45]
46	$\left[x+\frac{1}{2},\ z,\ y\right]$	[46]
47	$\left[\frac{1}{2}-z,\ y,\ -x\right]$	[47]
48	$\left[z+\frac{1}{2},\ y,\ x\right]$	[48]
49	$\left[x+\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	[49]
50	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}-z\right]$	[50]
51	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	[51]
52	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	[52]
53	$\left[z+\frac{1}{2},\ x,\ y+\frac{1}{2}\right]$	[53]
54	$\left[y+\frac{1}{2},\ z,\ x+\frac{1}{2}\right]$	[54]
55	$\left[\frac{1}{2}-y,\ z,\ \frac{1}{2}-x\right]$	[55]
56	$\left[\frac{1}{2}-z,\ -x,\ y+\frac{1}{2}\right]$	[56]
57	$\left[\frac{1}{2}-y,\ -z,\ x+\frac{1}{2}\right]$	[57]
58	$\left[z+\frac{1}{2},\ -x,\ \frac{1}{2}-y\right]$	[58]
59	$\left[y+\frac{1}{2},\ -z,\ \frac{1}{2}-x\right]$	[59]
60	$\left[\frac{1}{2}-z,\ x,\ \frac{1}{2}-y\right]$	[60]
61	$\left[-x,\ z+\frac{1}{2},\ -y\right]$	[61]
62	$\left[-x,\ \frac{1}{2}-z,\ y\right]$	[62]
63	$\left[-z,\ \frac{1}{2}-y,\ x\right]$	[63]
64	$\left[z,\ \frac{1}{2}-y,\ -x\right]$	[64]
65	$\left[y,\ \frac{1}{2}-x,\ -z\right]$	[65]
66	$\left[-y,\ x+\frac{1}{2},\ -z\right]$	[66]
67	$\left[-y,\ \frac{1}{2}-x,\ z\right]$	[67]
68	$\left[y,\ x+\frac{1}{2},\ z\right]$	[68]
69	$\left[x,\ \frac{1}{2}-z,\ -y\right]$	[69]
70	$\left[x,\ z+\frac{1}{2},\ y\right]$	[70]
71	$\left[-z,\ y+\frac{1}{2},\ -x\right]$	[71]
72	$\left[z,\ y+\frac{1}{2},\ x\right]$	[72]
73	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[73]
74	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[74]
75	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	[75]
76	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[76]
77	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ y\right]$	[77]
78	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ x\right]$	[78]
79	$\left[\frac{1}{2}-y,\ z+\frac{1}{2},\ -x\right]$	[79]

continued ...

Table 10

No.	position	mapping
80	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ y\right]$	[80]
81	$\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ x\right]$	[81]
82	$\left[z+\frac{1}{2},\ \frac{1}{2}-x,\ -y\right]$	[82]
83	$\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ -x\right]$	[83]
84	$\left[\frac{1}{2}-z,\ x+\frac{1}{2},\ -y\right]$	[84]
85	$\left[-x,\ z,\ \frac{1}{2}-y\right]$	[85]
86	$\left[-x,\ -z,\ y+\frac{1}{2}\right]$	[86]
87	$\left[-z,\ -y,\ x+\frac{1}{2}\right]$	[87]
88	$\left[z,\ -y,\ \frac{1}{2}-x\right]$	[88]
89	$\left[y,\ -x,\ \frac{1}{2}-z\right]$	[89]
90	$\left[-y,\ x,\ \frac{1}{2}-z\right]$	[90]
91	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	[91]
92	$\left[y,\ x,\ z+\frac{1}{2}\right]$	[92]
93	$\left[x,\ -z,\ \frac{1}{2}-y\right]$	[93]
94	$\left[x,\ z,\ y+\frac{1}{2}\right]$	[94]
95	$\left[-z,\ y,\ \frac{1}{2}-x\right]$	[95]
96	$\left[z,\ y,\ x+\frac{1}{2}\right]$	[96]
97	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[97]
98	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[98]
99	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	[99]
100	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[100]
101	$\left[z,\ x,\ y+\frac{1}{2}\right]$	[101]
102	$\left[y,\ z,\ x+\frac{1}{2}\right]$	[102]
103	$\left[-y,\ z,\ \frac{1}{2}-x\right]$	[103]
104	$\left[-z,\ -x,\ y+\frac{1}{2}\right]$	[104]
105	$\left[-y,\ -z,\ x+\frac{1}{2}\right]$	[105]
106	$\left[z,\ -x,\ \frac{1}{2}-y\right]$	[106]
107	$\left[y,\ -z,\ \frac{1}{2}-x\right]$	[107]
108	$\left[-z,\ x,\ \frac{1}{2}-y\right]$	[108]
109	$\left[\frac{1}{2}-x,\ z+\frac{1}{2},\ -y\right]$	[109]
110	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ y\right]$	[110]
111	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ x\right]$	[111]
112	$\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ -x\right]$	[112]
113	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ -z\right]$	[113]
114	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ -z\right]$	[114]
115	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z\right]$	[115]
116	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	[116]
117	$\left[x+\frac{1}{2},\ \frac{1}{2}-z,\ -y\right]$	[117]
118	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ y\right]$	[118]
119	$\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ -x\right]$	[119]
120	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ x\right]$	[120]
121	$\left[x,\ y+\frac{1}{2},\ z\right]$	[121]
122	$\left[x,\ \frac{1}{2}-y,\ -z\right]$	[122]
123	$\left[-x,\ y+\frac{1}{2},\ -z\right]$	[123]
124	$\left[-x,\ \frac{1}{2}-y,\ z\right]$	[124]
125	$\left[z,\ x+\frac{1}{2},\ y\right]$	[125]
126	$\left[y,\ z+\frac{1}{2},\ x\right]$	[126]

continued ...

Table 10

No.	position	mapping
127	$\left[-y,\, z+\frac{1}{2},\, -x\right]$	[127]
128	$\left[-z,\, \frac{1}{2}-x,\, y\right]$	[128]
129	$\left[-y,\, \frac{1}{2}-z,\, x\right]$	[129]
130	$\left[z,\, \frac{1}{2}-x,\, -y\right]$	[130]
131	$\left[y,\, \frac{1}{2}-z,\, -x\right]$	[131]
132	$\left[-z,\, x+\frac{1}{2},\, -y\right]$	[132]
133	$\left[\frac{1}{2}-x,\, z,\, \frac{1}{2}-y\right]$	[133]
134	$\left[\frac{1}{2}-x,\, -z,\, y+\frac{1}{2}\right]$	[134]
135	$\left[\frac{1}{2}-z,\, -y,\, x+\frac{1}{2}\right]$	[135]
136	$\left[z+\frac{1}{2},\, -y,\, \frac{1}{2}-x\right]$	[136]
137	$\left[y+\frac{1}{2},\, -x,\, \frac{1}{2}-z\right]$	[137]
138	$\left[\frac{1}{2}-y,\, x,\, \frac{1}{2}-z\right]$	[138]
139	$\left[\frac{1}{2}-y,\, -x,\, z+\frac{1}{2}\right]$	[139]
140	$\left[y+\frac{1}{2},\, x,\, z+\frac{1}{2}\right]$	[140]
141	$\left[x+\frac{1}{2},\, -z,\, \frac{1}{2}-y\right]$	[141]
142	$\left[x+\frac{1}{2},\, z,\, y+\frac{1}{2}\right]$	[142]
143	$\left[\frac{1}{2}-z,\, y,\, \frac{1}{2}-x\right]$	[143]
144	$\left[z+\frac{1}{2},\, y,\, x+\frac{1}{2}\right]$	[144]
145	$\left[x+\frac{1}{2},\, y,\, z\right]$	[145]
146	$\left[x+\frac{1}{2},\, -y,\, -z\right]$	[146]
147	$\left[\frac{1}{2}-x,\, y,\, -z\right]$	[147]
148	$\left[\frac{1}{2}-x,\, -y,\, z\right]$	[148]
149	$\left[z+\frac{1}{2},\, x,\, y\right]$	[149]
150	$\left[y+\frac{1}{2},\, z,\, x\right]$	[150]
151	$\left[\frac{1}{2}-y,\, z,\, -x\right]$	[151]
152	$\left[\frac{1}{2}-z,\, -x,\, y\right]$	[152]
153	$\left[\frac{1}{2}-y,\, -z,\, x\right]$	[153]
154	$\left[z+\frac{1}{2},\, -x,\, -y\right]$	[154]
155	$\left[y+\frac{1}{2},\, -z,\, -x\right]$	[155]
156	$\left[\frac{1}{2}-z,\, x,\, -y\right]$	[156]
157	$\left[-x,\, z+\frac{1}{2},\, \frac{1}{2}-y\right]$	[157]
158	$\left[-x,\, \frac{1}{2}-z,\, y+\frac{1}{2}\right]$	[158]
159	$\left[-z,\, \frac{1}{2}-y,\, x+\frac{1}{2}\right]$	[159]
160	$\left[z,\, \frac{1}{2}-y,\, \frac{1}{2}-x\right]$	[160]
161	$\left[y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[161]
162	$\left[-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[162]
163	$\left[-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[163]
164	$\left[y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[164]
165	$\left[x,\, \frac{1}{2}-z,\, \frac{1}{2}-y\right]$	[165]
166	$\left[x,\, z+\frac{1}{2},\, y+\frac{1}{2}\right]$	[166]
167	$\left[-z,\, y+\frac{1}{2},\, \frac{1}{2}-x\right]$	[167]
168	$\left[z,\, y+\frac{1}{2},\, x+\frac{1}{2}\right]$	[168]
169	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[169]
170	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[170]
171	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[171]
172	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[172]
173	$\left[z+\frac{1}{2},\, x+\frac{1}{2},\, y+\frac{1}{2}\right]$	[173]

continued ...

Table 10

No.	position	mapping
174	$\left[y+\frac{1}{2},\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$	[174]
175	$\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, \frac{1}{2}-x\right]$	[175]
176	$\left[\frac{1}{2}-z,\, \frac{1}{2}-x,\, y+\frac{1}{2}\right]$	[176]
177	$\left[\frac{1}{2}-y,\, \frac{1}{2}-z,\, x+\frac{1}{2}\right]$	[177]
178	$\left[z+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-y\right]$	[178]
179	$\left[y+\frac{1}{2},\, \frac{1}{2}-z,\, \frac{1}{2}-x\right]$	[179]
180	$\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, \frac{1}{2}-y\right]$	[180]
181	$[-x,\, z,\, -y]$	[181]
182	$[-x,\, -z,\, y]$	[182]
183	$[-z,\, -y,\, x]$	[183]
184	$[z,\, -y,\, -x]$	[184]
185	$[y,\, -x,\, -z]$	[185]
186	$[-y,\, x,\, -z]$	[186]
187	$[-y,\, -x,\, z]$	[187]
188	$[y,\, x,\, z]$	[188]
189	$[x,\, -z,\, -y]$	[189]
190	$[x,\, z,\, y]$	[190]
191	$[-z,\, y,\, -x]$	[191]
192	$[z,\, y,\, x]$	[192]